windcheck

Label-free detection of cross-wrap self-coincidence
in Herculaneum scroll segmentations

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Summary

A traced papyrus surface that goes once around the scroll should come back onto the *next* sheet. `windcheck` finds the places where it comes back onto the *same* sheet instead.

The check runs on the published `tifxyz` meshes alone. It needs no ground-truth labels, no annotations, no trained model, no GPU, and no volume download: a full scroll of 53 surfaces is analysed in a few minutes on a laptop from about 670 MB of surface data. It ships with a calibration command, a null control that passes, and a validity filter that refuses to report rows it cannot stand behind.

This addresses Open Problems §2 and §3 — *"sheet switches, where a traced mesh jumps from the intended sheet to a neighboring wrap"* and mesh topology repair, described there as *"one of the highest-leverage places to contribute."*

1 The problem

A scroll is one sheet, rolled. Segmentation traces that sheet through the CT volume. When a tracer crosses to an adjacent wrap, the output still looks like a clean surface, and the text rendered from it still looks like text — but it is spliced from two different parts of the scroll. The failure is silent, which is what makes it expensive.

There is no published set of known-bad segments, so a supervised detector cannot be trained and a supervised evaluation cannot be run. Any usable check must therefore be derived from a property the correct answer must satisfy, and must carry its own controls.

2 Method

A `tifxyz` surface is a 2D grid of 3D vertex coordinates: grid index (v, u) maps to a volume coordinate. Walking along u goes around the scroll.

For every grid point p, measure the distance to the nearest point of *the same surface* that is at least E grid columns away in u:

$$g(p) \;=\; \min_{|u_q - u_p| \geq E} \|p - q\|, \qquad q \in S.$$

The exclusion removes the patch of surface p already sits on, so what remains is the trace's own neighbouring wrap. A correct trace reads one sheet spacing there. A value near zero means the trace is lying on top of a wrap it has already traced.

Two properties of this definition matter.

It never needs a scroll axis. An earlier version of this work measured radius against winding number instead. That approach fails its own positive control: fitting radius on the 44 labelled sheets of Scroll 5 — ground truth, known radial order — leaves a residual standard deviation of 79.8 voxels against a sheet spacing of 12–17 voxels, because a carbonised scroll is crushed, not round. Comparing a trace to itself is invariant to that deformation.

Distance is measured to the interpolated surface, not to the nearest grid sample. The grids are sampled roughly every 20 voxels while adjacent sheets sit about 17 voxels apart, so a nearest-sample method has less than a 1.5× margin and cannot separate one sheet of gap from two.

The kernel is C++17 with a uniform-grid acceleration structure and a `std::thread` pool, sustaining roughly 27,000 point-to-surface queries per second on 12 threads against a 30-million-triangle atlas.

3 Calibration

Thresholds are meaningless until the scroll says what a sheet is worth. The `calibrate` command measures the distance from each labelled winding to the sheet k windings away.

separation	surface pairs	median distance	ratio
1 sheet	43	17.05 vx	1.000
2 sheets	42	34.14 vx	2.003
3 sheets	41	51.22 vx	3.004

Table 1: Sheet separation on `PHerc0172`, over 126 surface pairs. The linearity is what licenses reading a doubled gap as a skipped wrap. If it does not hold on a given scroll, the labels are not a radial index and the check does not apply there.

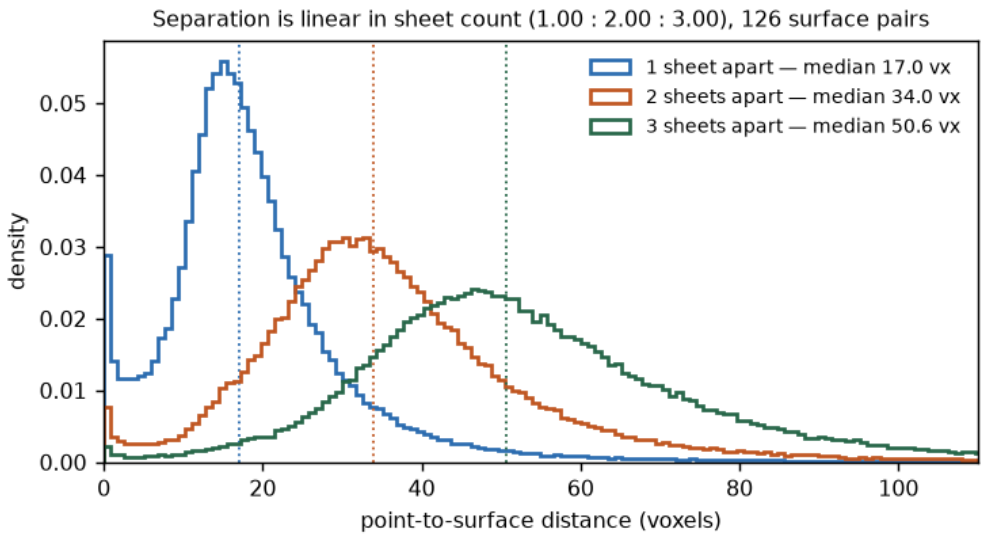

Figure 1: The same measurement as distributions. The overlap is shown deliberately: per-*point* discrimination between a one- and two-sheet gap is only about 78%, which is why every verdict in this work is made over a region and never over a single point.

4 Controls

Null control. A single-wrap segment has no previous wrap, so a working check must find nothing on it. All 44 labelled single-winding segments of Scroll 5 report 0.00% flagged. The same property is pinned by a unit test on a synthetic flat sheet, so it cannot silently regress.

Validity filter. Exclusion by u-index assumes u advances with angle. Where that assumption is weak, a flagged partner may be a same-wrap neighbour rather than the next wrap. Any trace whose flagged $|\Delta u|$ tenth percentile falls within $2E$ is rejected — in code, not by inspection.

The filter is not decorative. On `PHerc0814` it rejects the two *highest-scoring* traces (11.27% and 8.18% of points inside the 2.0-voxel threshold), both artifacts of the exclusion cutoff. Without it, they would have been the headline numbers of this submission.

Cross-check. On the traces that pass, flagged partners sit at $|\Delta u| = 271$–618 columns, matching revolution periods derived independently by angular unwrapping (236–616) across all nine traces.

5 Results

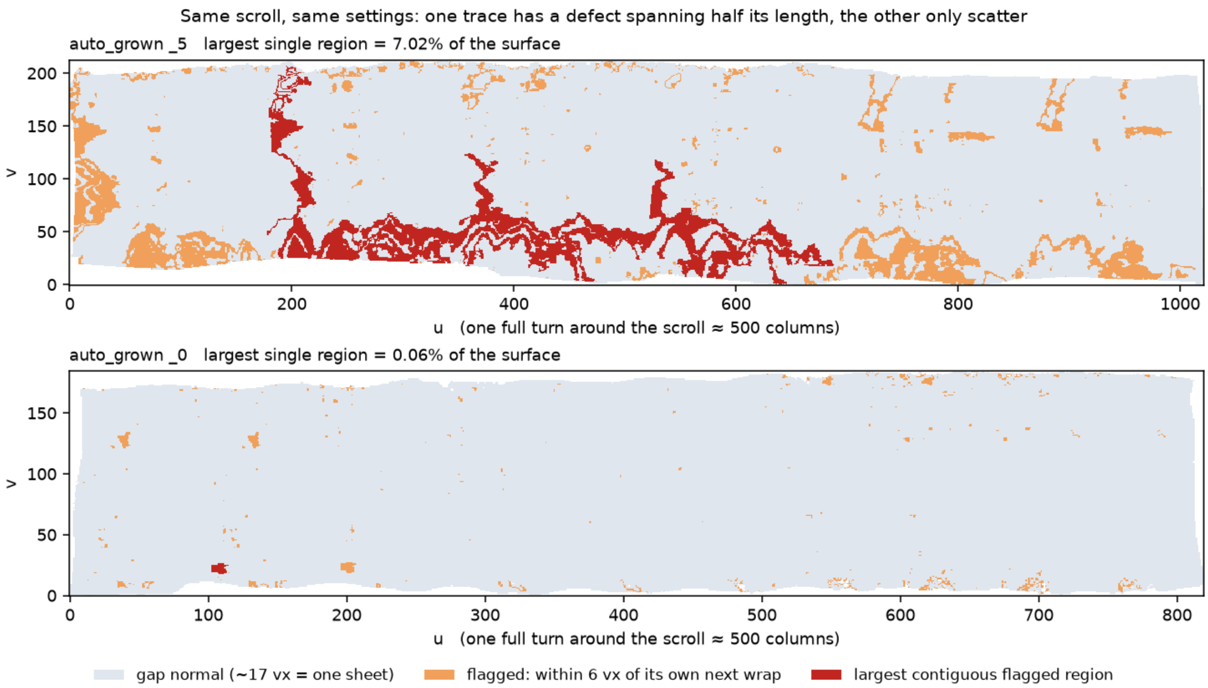

Figure 2: Two traces from the same scroll under identical settings. The upper has a single contiguous flagged region covering 7.02% of its surface and spanning roughly half its length; the lower has only scatter.

Across 19 analysed traces on two scrolls, two facts separate cleanly.

Scattered flagging is normal. The overall violation rate overlaps between scrolls (`PHerc0172` 0.36–6.60%, `PHerc0814` 0.32–6.77%). Taken alone it distinguishes nothing, and should not be read as a defect signature.

Concentration does not. Measured as the largest contiguous flagged region as a fraction of analysed surface, one trace stands at 7.02% against 1.52% for the next and 0.06% for the lowest. That ordering is the usable output: it says which trace a human should open first.

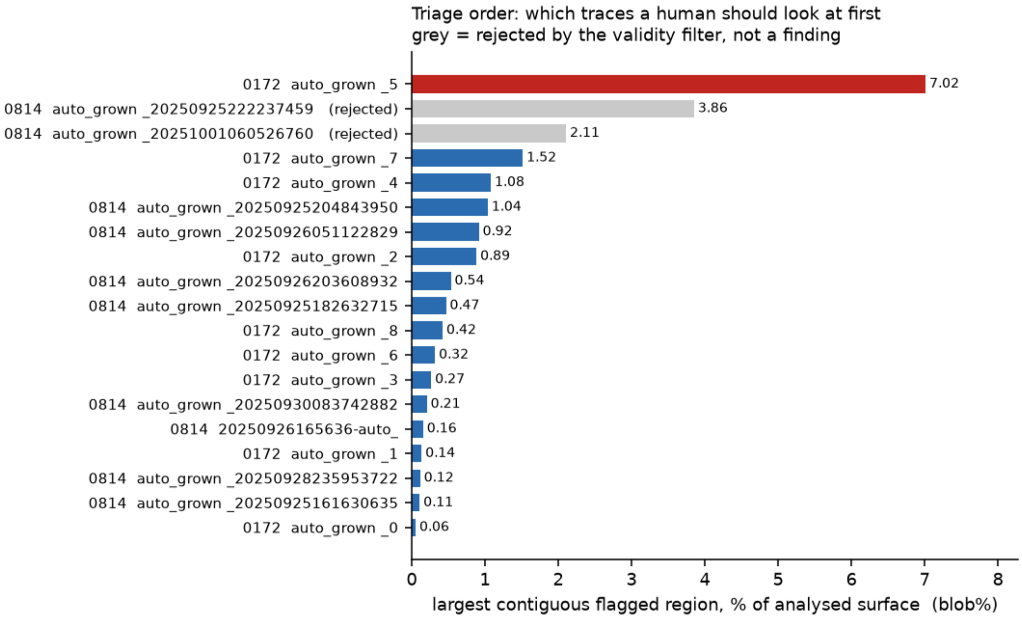

Figure 3: All 19 analysed traces ordered by concentration. Rows rejected by the validity filter are greyed rather than removed.

6 What this does not establish

Stated plainly, because stronger versions of each claim were tested and dropped.

These regions are not proven to be errors. Three independent attempts to confirm them against the CT volume failed. The strongest, a missed-sheet test asking whether a nearby bright band is occupied by no part of the trace, gave Cohen's $d = 0.245$ at $p = 0.0496$ — in the predicted direction, but far too small to certify any individual region. The geometry is solid; the interpretation as a defect is not established.

No claim is made that the published output violates the criterion of the grower that produced it. `volume-cartographer` defines `same_surface_th = 2.0` and rejects growth candidates within that distance of already-traced surface, which is why 2.0 voxels is reported as a column. But the published Scroll 5 meshes pass through private artifacts, flattening and reconstruction before release, so their provenance to any specific revision is not established. What can be said is narrower and still useful: no stage re-checks that criterion after optimisation.

The sample is small. The nine Scroll 5 `auto_grown` traces are partitions of a single artifact, not nine independent runs. The effective sample is one run plus eight `PHerc0814` segments. Two scrolls is not enough to characterise a pipeline.

Known limitations. The exclusion width is fixed rather than adapted to each trace's turn period; exclusion is by grid column rather than mesh geodesic distance; and single-wrap patches cannot be analysed at all.

7 Reproducing

Everything below runs from published data with no credentials.

```
uv sync
clang++ -O3 -std=c++17 -pthread \
    -o engines/atlas_query engines/atlas_query.cpp
```

```
uv run pytest -q

uv run python -m windcheck.fetch --sample PHerc0172
uv run windcheck calibrate data/scroll5_tifxyz
uv run windcheck selfgap   data/scroll5_tifxyz --json report.json
```

`data/MANIFEST.json` pins every input file by S3 key, byte count and SHA-256, so a result can be traced to exact bytes. The outputs quoted in this document are committed under `sample_outputs/`, each headed by the command that produced it.

Five tests run without any data. Each is tied to a mistake made during development: the accelerated search against brute force; certification of the search radius; true surface separation against grid pitch; the null control; and missing-cell handling.

Availability

Source, figures and sample outputs: `https://github.com/joe-carr-data/windcheck`